Problem 1

$$\int_0^1 x^{44} \ln(x) \, \mathrm{d}x$$

Problem 2

$$\int_0^{\frac{\pi}{4}} (1 - x \tan(x)) \tan(x) \, \mathrm{d}x$$

Problem 3

$$\int_{-1}^1 f_n(x) f_m(x) \, \mathrm{d}x \qquad \text{for positive integers } m, n,$$

where $f_n(x) := \sum_{k=0}^{n+2025} \left(x + \sqrt{x^2 - 1} \cos\left(\frac{2\pi k}{n + 2026} \right) \right)^{n+26}$.

Problem 4

$$\int_0^\infty \int_0^\infty \sin\left(x^2 + y^2\right) \, \mathrm{d}x \, \mathrm{d}y$$

Problem 5

$$\int_0^1 \frac{(\ln(1-x^n))^m}{x}\,\mathrm{d}x \qquad \text{for positive integers } m, n$$

Problem 6

$$\int \frac{f^{-1}(x)}{1+f(x)}\,\mathrm{d}x\,,$$

where $f(x) := x^{\varphi}$, $\varphi = \dfrac{1+\sqrt{5}}{2}$ is the golden ratio.

Problem 7

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin(x)^{\tan(x)-2}\left(\tan^2(x)\ln(\sin(x)) + \sin^2(x)\right)\mathrm{d}x$$

Problem 8

$$\int_0^{2\pi} \frac{\sin^{67}(x)}{2026 - \cos(x)} \, \mathrm{d}x$$

Problem 9

$$\int_0^{\infty} \frac{\sin(x)}{x(\cosh(x) + \cos(x))} \, \mathrm{d}x$$

Problem 10

$$\prod_x (2025x + 2026),$$

where $\prod_x$ is the discrete multiplicative integral, i.e. an inverse operator of $g(f(x)) = \dfrac{f(x+1)}{f(x)}$.

Problem 11

$$\int_0^{\infty} \frac{\sin(x)\ln(x)}{x} \, \mathrm{d}x$$

Problem 12

$$\int_{-1}^{1} \frac{1+\mathrm{Li}_2(x)}{4+\mathrm{Li}_2(x^2)}\,\mathrm{d}x\,,$$

where $\mathrm{Li}_2(x) := -\int_0^x \frac{\ln(1-t)}{t}\,\mathrm{d}t$ is the dilogarithm.

Problem 13

$$\int_{-\infty}^{\infty} e^{x-\sinh^2(x)}\,\mathrm{d}x$$

Problem 14

$$\int_0^{\infty} \frac{x^{\frac{e}{2}-1}(1+\ln(x))}{x^e+1}\,\mathrm{d}x$$

Problem 15

$$\int_0^1 \frac{1-x}{1+x}\cdot\frac{1}{\sqrt{x+x^2+x^3}}\,\mathrm{d}x$$

Problem 16

$$\int_{\sqrt{2}-1}^{\sqrt{2}+1} \frac{1}{x\left(e+\left(x^2+1\right)^{\frac{1}{\ln(x)}}\right)}\,\mathrm{d}x$$

Problem 17

$$\int_{1}^{\infty} \frac{1}{\lfloor x+\lfloor x\rfloor\rfloor^2}\,\mathrm{d}x$$

Problem 18

$$\int_{0}^{\infty} \frac{x^2-1}{x^2+1}\cdot\frac{\sin(x)}{x}\,\mathrm{d}x$$

Problem 19

$$\lim_{n\to\infty} n\int_{0}^{1} \frac{1+x^3}{n+x+x^3+x^5}\,\mathrm{d}x$$

Problem 20

$$\left\lfloor \int_2^{2025} \frac{1}{\lfloor 2\log_x(5) + 4\log_x(3) \rfloor} \, \mathrm{d}x \right\rfloor$$